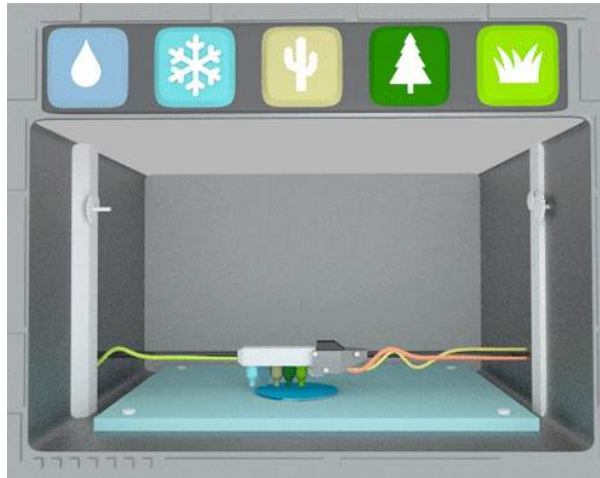




Source: 3D Printing World Environment Day GIF By General Electric

Embedding Sustainability in Design Optimization: Which Mathematical Recipes?

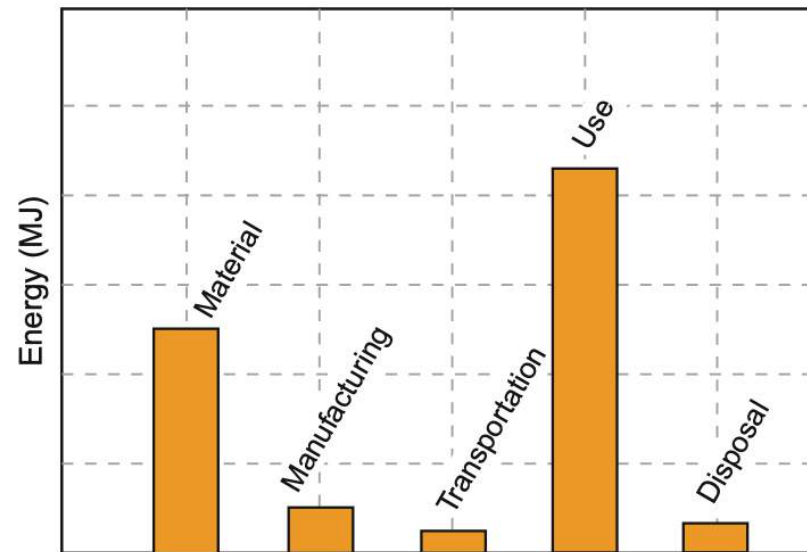
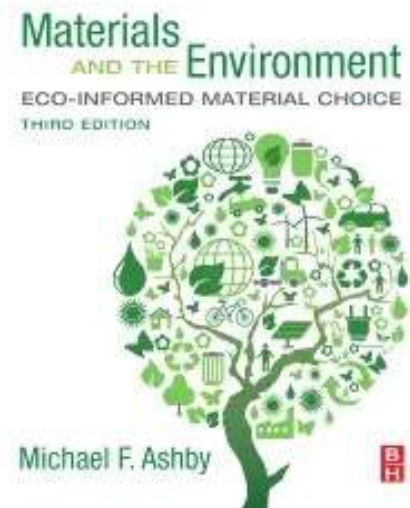
Prof. Joseph Morlier



Thanks to all my
PhDs, MsCs and
co-authors



Environnemental Footprint

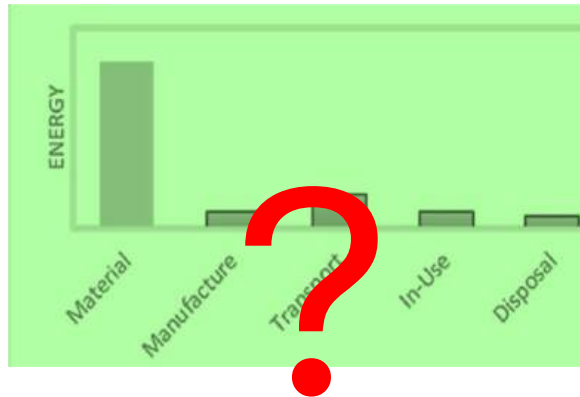
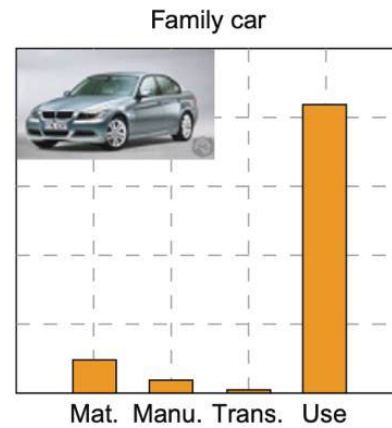
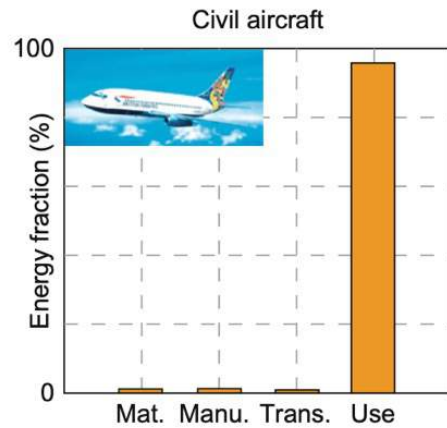


$\propto CO_2$

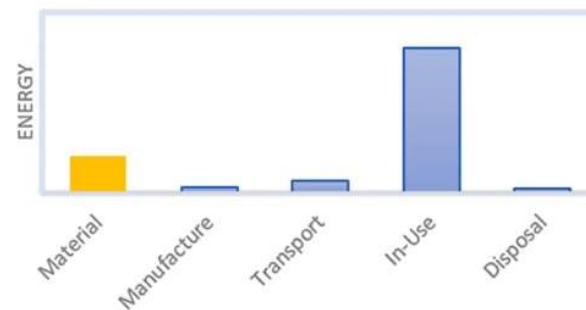
Breakdown of energy into that associated with each life phase



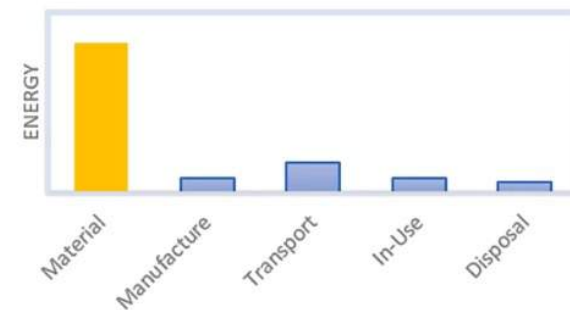
Electrification example (from automotive)



Energy & CO₂ over vehicle life
Pre-Electrification



Energy & CO₂ over vehicle life
Post-Electrification



<https://www.ansys.com/blog/the-impact-of-materials-on-sustainability-part-2>

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Au programme

1. Problem $\mathcal{P}_1$, where the topology is fixed (truss) or parametric (wing),
2. Problem $\mathcal{P}_2$, where the topology is free.
3. On some coupled problems $\mathcal{P}_3$
4. Conclusions

Au programme

- 1. Problem $\mathcal{P}_1$, where the topology is fixed (truss) or imposed (wing),**
2. Problem $\mathcal{P}_2$, where the topology is free.
3. On some coupled problems $\mathcal{P}_3$
4. Conclusions

$$\mathcal{P}_1$$

- Continuous relaxation *versus* Variational Autoencoders (VAEs)

ECO Hale

<https://www.openscience.fr/Maitriser-la-conception-des-drones-solaires-a-voilure-souple-vers-l-avenement>

Derive aerostructures MDO tools to treat a HALE pseudo satellite Design problem

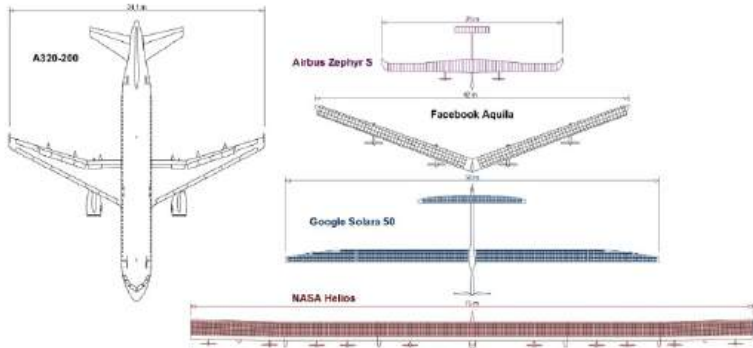
www.nature.com/scientificreports

scientific reports

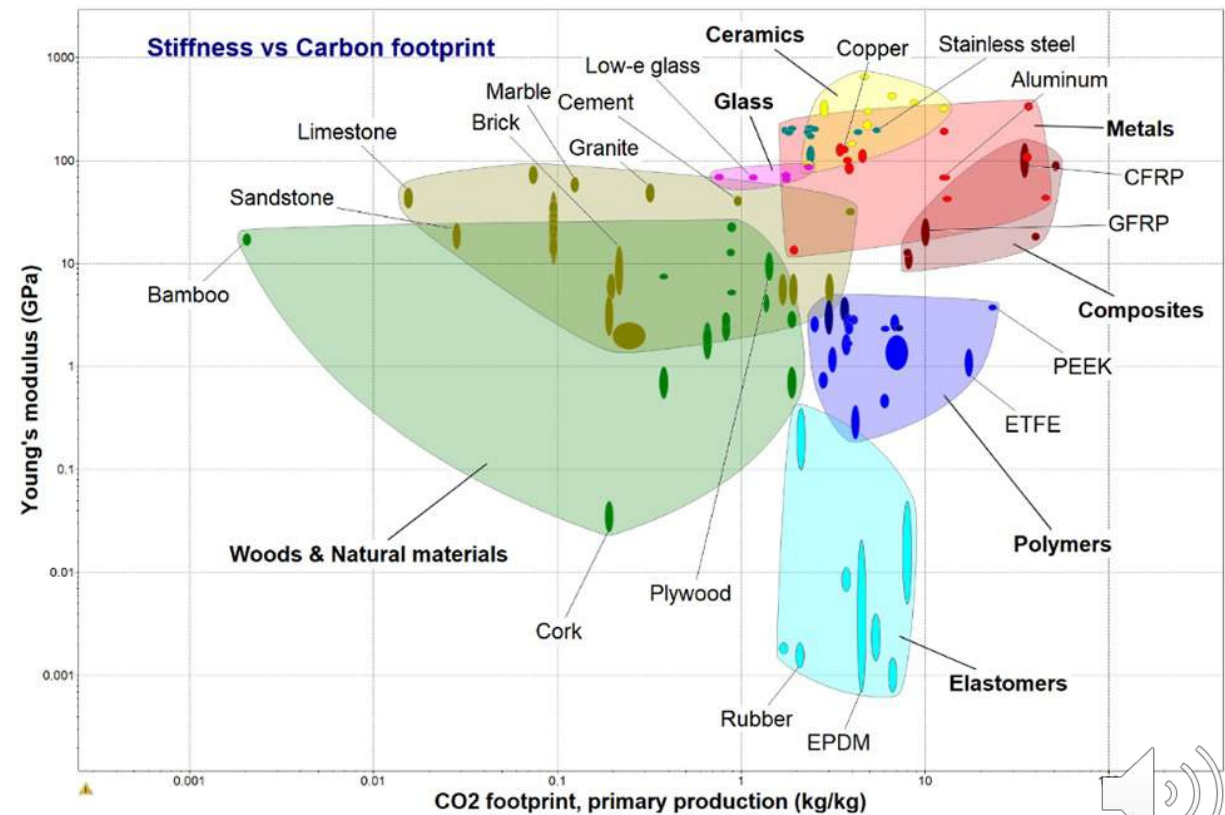
OPEN CO₂ footprint minimization of solar-powered HALE using MDO and eco-material selection

Edouard Duriez^{1,3}, Víctor Manuel Guadaño Martín^{2,3} & Joseph Morlier^{1,3}

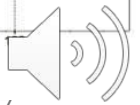
Assets: Flexible, repositionable, permanent cover cheaper, lower environmental impact?



Dis



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ECO MATERIAL SELECTION

Turn discrete problem of material selection into continuous inspired from SIMP

Zuo, W., & Saitou, K. (2016). Multi-material topology optimization using ordered SIMP interpolation. Structural and Multidisciplinary Optimization, 55(2), 477-491. doi:10.1007/s00158-016-1513-3

- Access material properties through density (only one material design variable) :

$E(\rho)$, $G(\rho)$, $\sigma_{\text{yield}}(\rho)$, $\text{CO}_2(\rho)$ (production CO2 footprint)

- Example for young modulus : Between two materials i and $i+1$,

$$E_e(\rho_e) = A_E * \rho_e^p + B_E, \text{ with } A_E = \frac{E_i - E_{i+1}}{\rho_i^p - \rho_{i+1}^p} \text{ and } B_E = E_i - A_E * \rho_i^p$$

- Power p added during optimisation so that only real materials are optimal
- Inversed curvature for CO2 as smaller CO2 is more advantageous

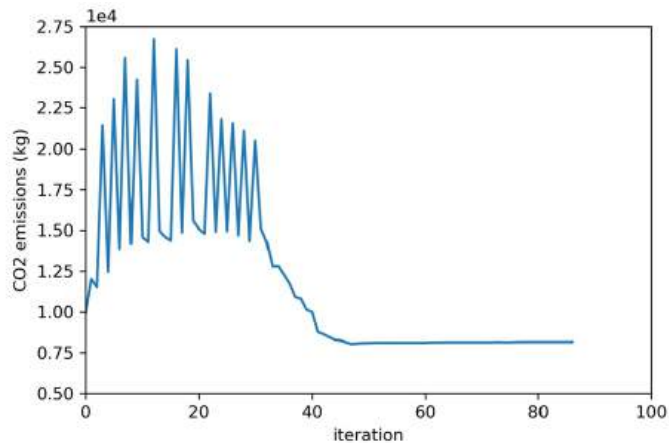
Our design variables

wing **topology** and **materials** !

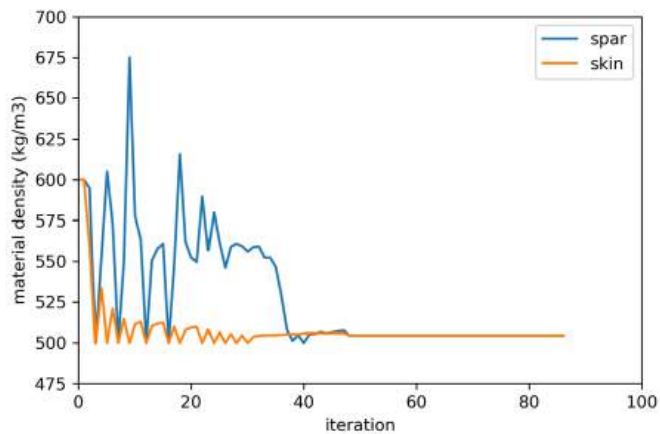
Variable	Lb	Ub	X0 (multistart)	unit
Skin thickness	0,0001	0,1	0,01-0,001	m
Spar thickness	0,0001	0,1	0,01-0,001	m
Wing span	1	1000	50	m
Wing chord	1	500	2	m
Wing taper	0,01	0,99	0,1	-
Wing thickness over chord ratio	0,01	0,20	0,1	-
Twist	-15	15	5	°
Density	50	8200	50-8200	Kg/m3

CO2 footprint minimization

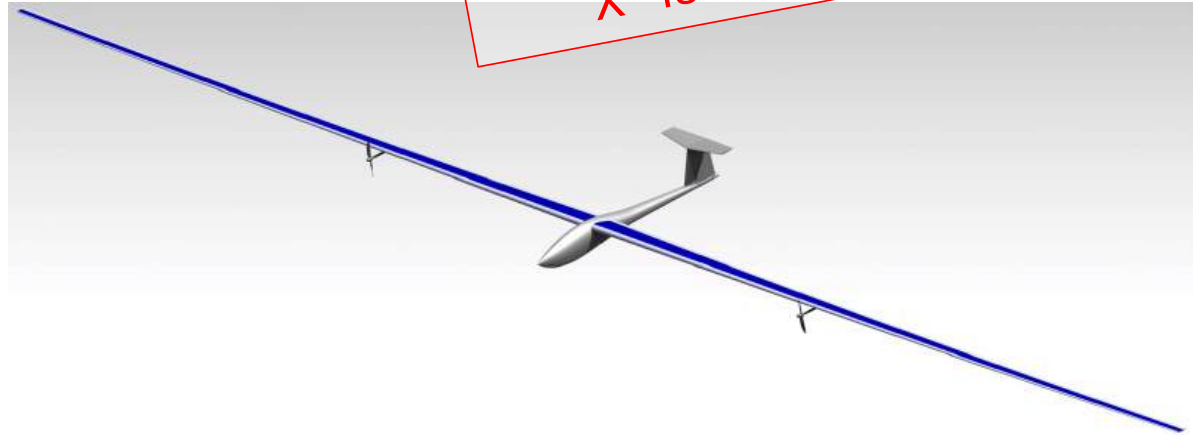
(a) Objective function: total CO2 emitted:



(b) Material density for skins and spars:



Convergence graphs



CAD model of the optimal HALE obtained

A slight increase in the total weight of the drone leads to an increase in the weight of the battery and the solar panel in order to propel a heavier drone,

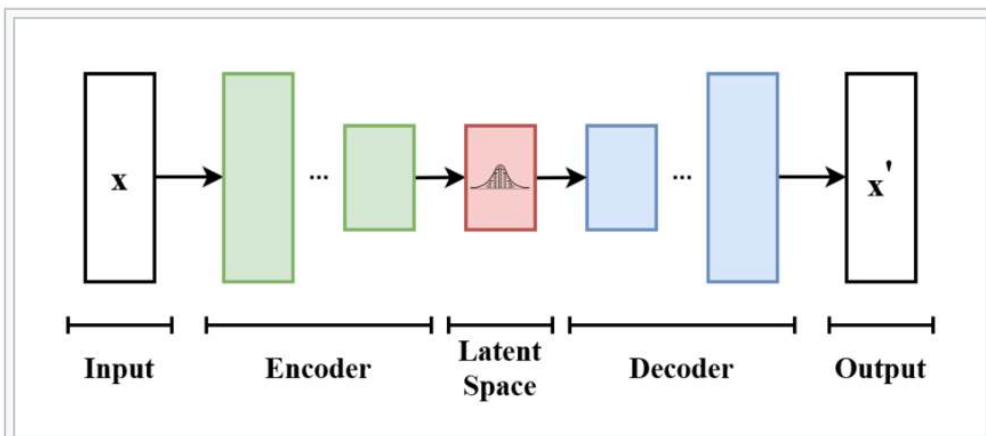
But also to: an increase in the weight of the wing structure that induces a more important lift to compensate → increase in the overall weight of the drone.

→ “snowball” effect.



The main problem: introducing *discrete* material database in a *continuous* gradient based optimization

the environmental choices most often rely on database of discrete properties (Ansys Granta, Ecoinvent etc...).



The basic scheme of a variational autoencoder. The model receives x as input. The encoder compresses it into the latent space. The decoder receives as input the information sampled from the latent space and produces x' as similar as possible to x .

HOW?

Variational AutoEncoder (VAE):
class of probabilistic models that find latent, low-dimensional representations of data.

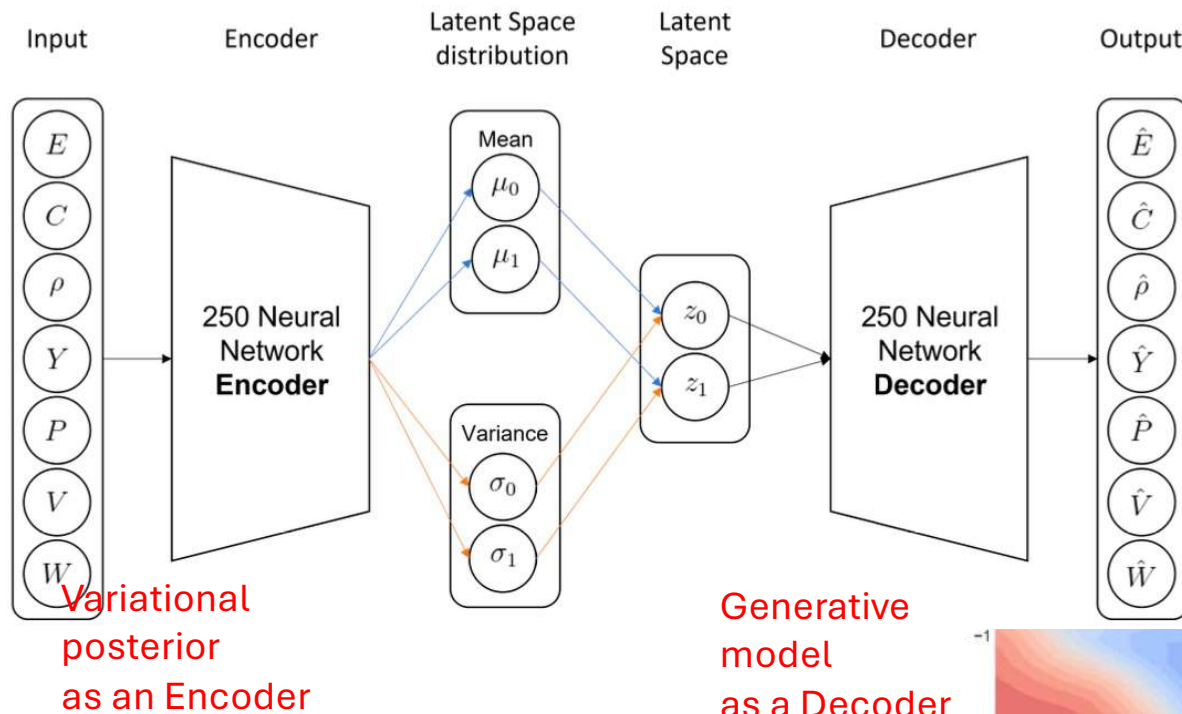
Kingma, D. P., & Welling, M. "Auto-Encoding Variational Bayes", 2013.



From catalog to *differentiable* material representation

Table 1 A curated subset of materials and their training

Material	Class	E [Pa]	Cost C [\$/kg]
A286 iron	Steel	2.01E+11	5.18E+00
AISI 304	Steel	1.90E+11	2.40E+00
Gray cast iron	Steel	6.62E+10	6.48E-01
3003-H16	Al alloy	6.90E+10	2.18E+00
5052-O	Al alloy	7.00E+10	2.23E+00
7050-T7651	Al alloy	7.20E+10	2.33E+00
Acrylic	Plastic	3.00E+09	2.80E+00
ABS	Plastic	2.00E+09	2.91E+00
PE HD	Plastic	1.07E+09	2.21E+00



Framework: PyTorch VAE

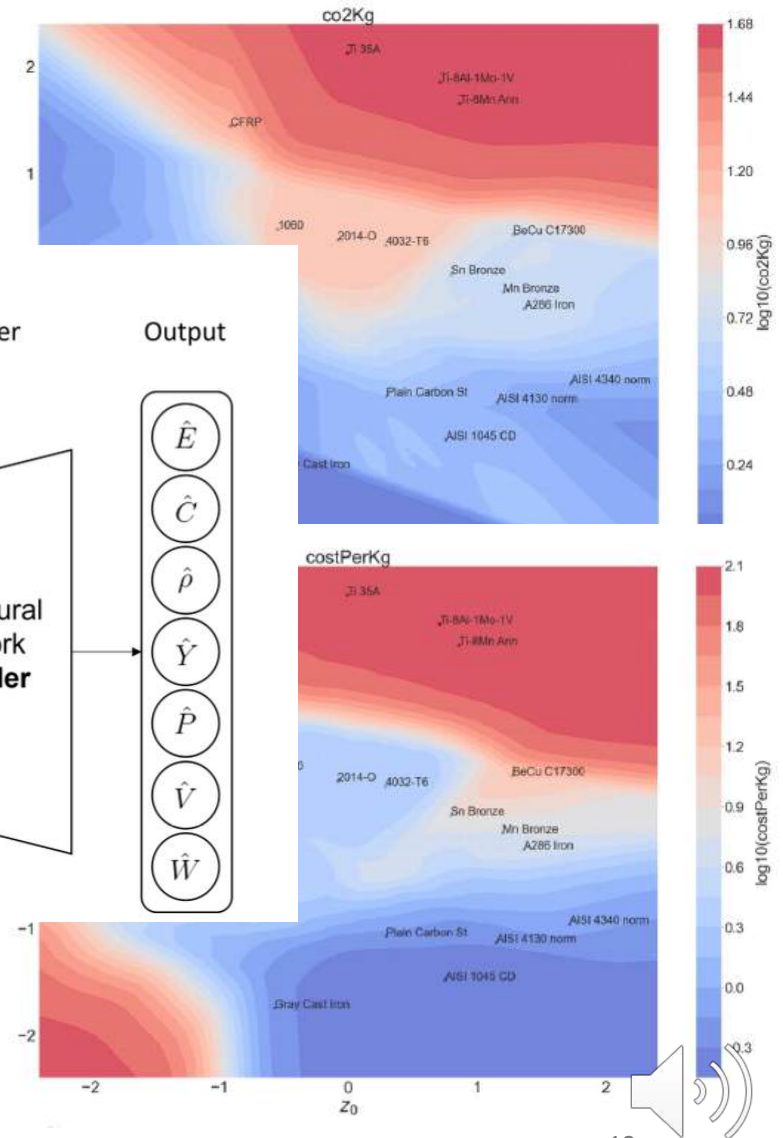
Network architecture: Fully connected with 250 neurons in encoder and decoder

Latent space: 2D (z_0, z_1)

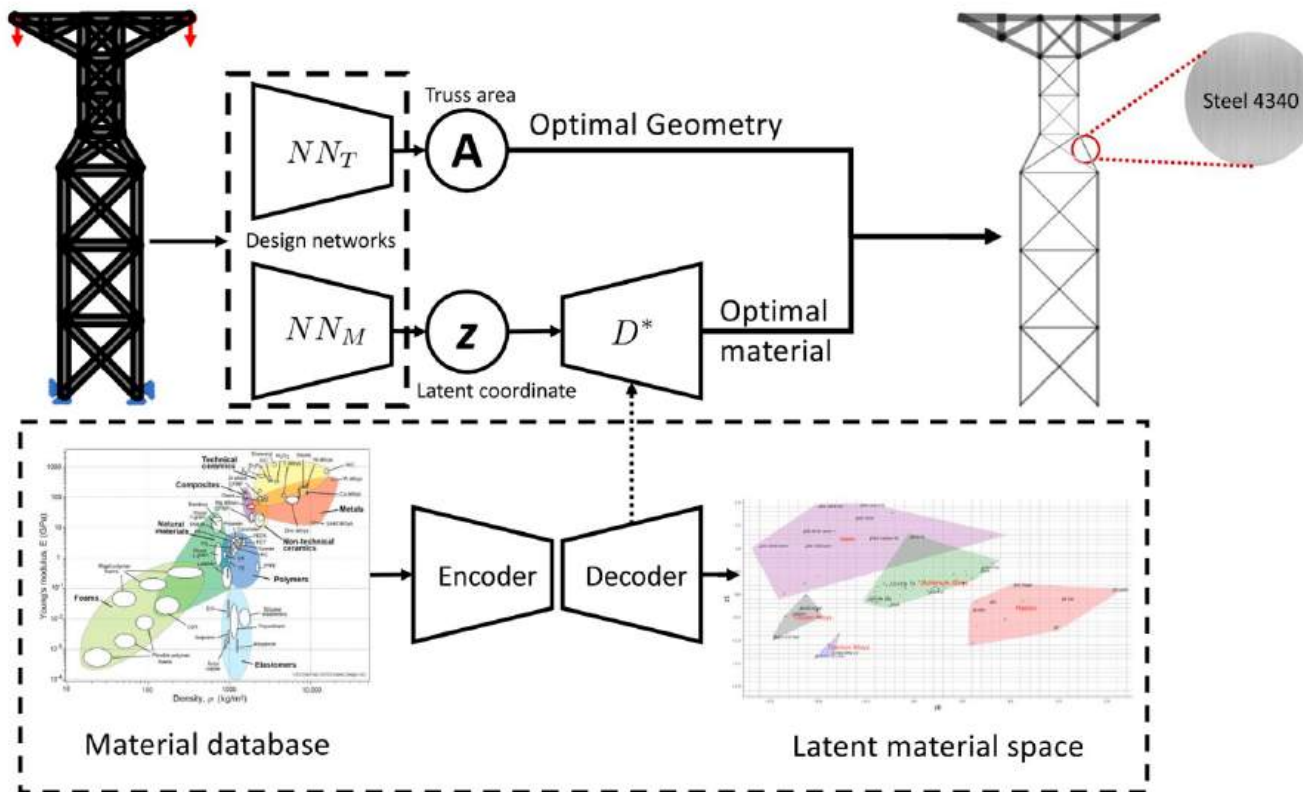
Inputs normalized to $[0, 1]$; outputs rescaled post-decoding

Training time: 3 minutes on a CPU (50,000 iterations)

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Everything is continuous here !



Chandrasekhar, A., Sridhara, S., & Suresh, K. (2022). Integrating material selection with design optimization via neural networks. *Engineering with Computers*, 38(5), 4715-4730.

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LCA & eco selection

- Material
- Process
- from cradle to grave
- ...

Water withdrawal



Generation of waste



Use recycled:
Fibers
Resin
Metals
Reuse & Repair

Carbon Footprint



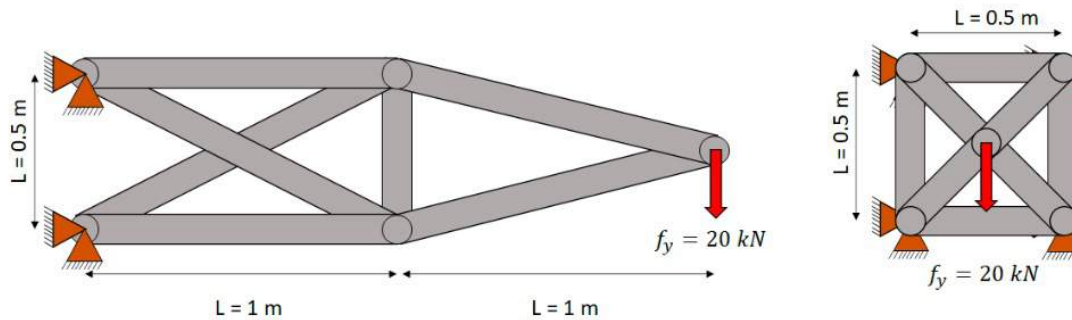
Energy requirement



Yepes Llorente, L., Morlier, J., Sridhara, S., & Suresh, K. (2024). A hybrid machine learning and evolutionary approach to material selection and design optimization for eco-friendly structures. *Structural and Multidisciplinary Optimization*, 67(5), 69.



Environmentally friendly 3D truss

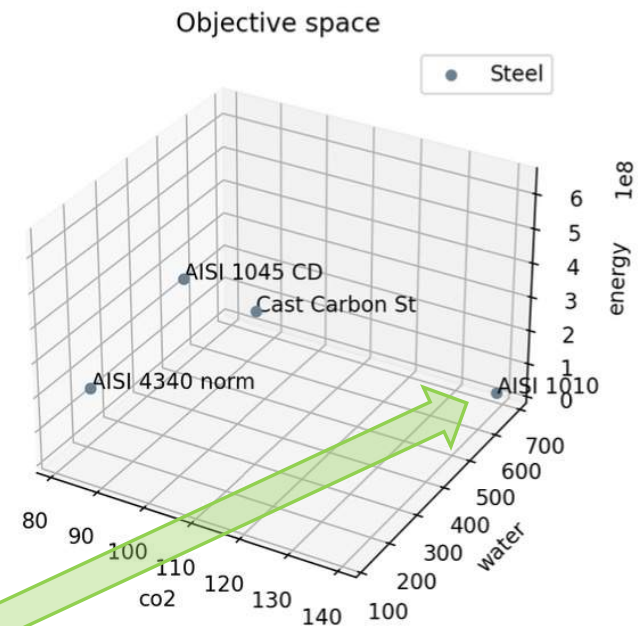


Being P_{min} , W_{min} and V_{min} the minimum value of CO_2 , water and energy achievable respectively, the index is defined as:

$$H = \frac{1}{3} \frac{P - P_{min}}{P_{min}} + \frac{1}{3} \frac{W - W_{min}}{W_{min}} + \frac{1}{3} \frac{V - V_{min}}{V_{min}} \quad (7)$$

In such a way that the perfect material with minimum consumption would yield $H = 0$. To take into account the cost of the structure as well, both values were calculated for each solution and presented in Table 10.

Material	Cost	Env. Consumption
AISI 1045 CD	18.63 \$	9.60 [-]
Cast Carbon St	28.68 \$	7.85 [-]
AISI 4340 norm	31.91 \$	2.28 [-]
AISI 1010	24.04 \$	1.73 [-]



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Au programme

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4. Conclusions

$$\mathcal{P}_2$$

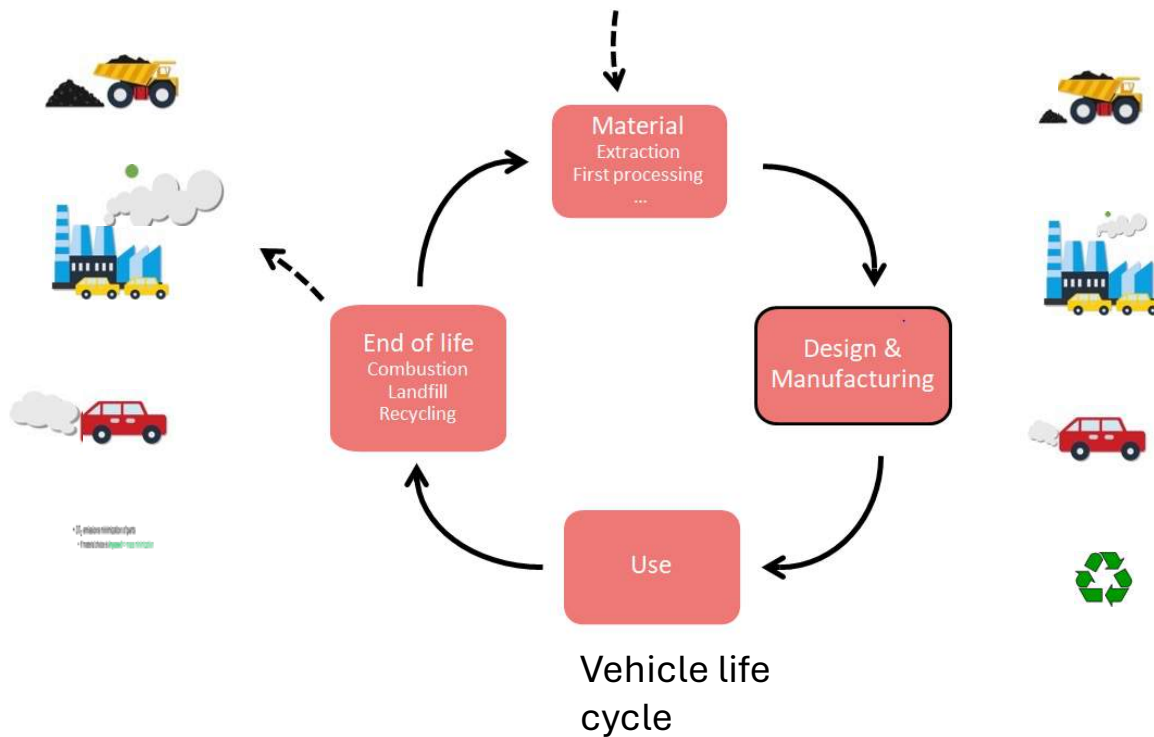
$$\begin{array}{ll} \min_{m,p,\mathcal{D}} & CO_2^{\text{tot}}(\mathcal{D}, m, p) \\ \text{s.t.} & \delta \leq \delta_{\text{max}} \\ & \{m, p\} \in \Phi_{\text{material}} \times \Phi_{\text{process}} \end{array}$$

- Topology, material and process.



Hypothesis 1

- CO_2 emissions minimization of parts
 - If material choice is **imposed** => mass minimization

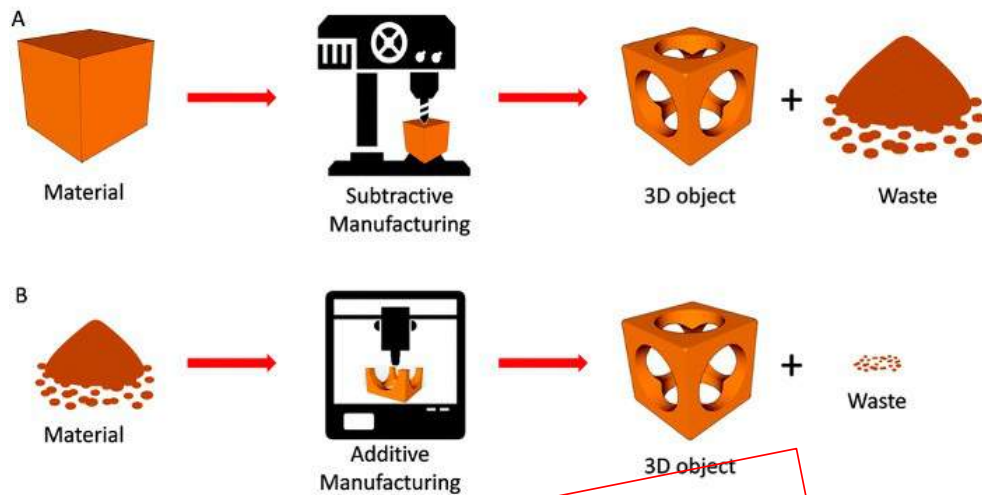


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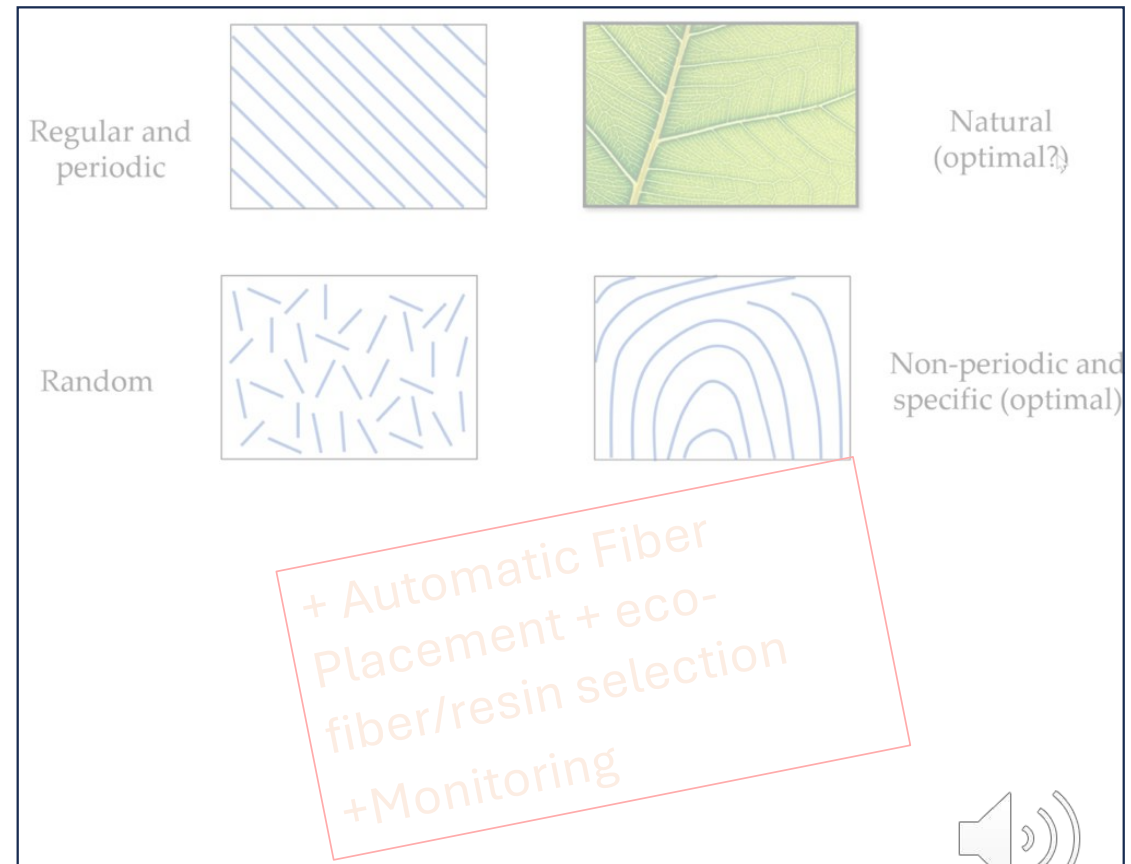


Process is AM, but WHY?

<https://dfam.substack.com/p/dfam-education-in-2022>

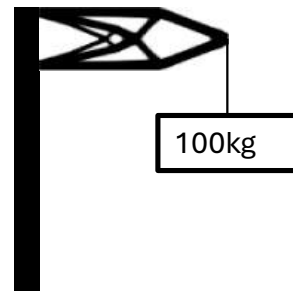
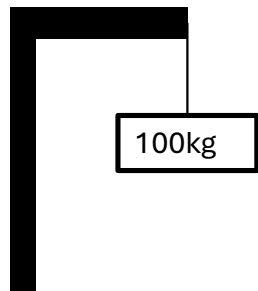


+Near 100% material utilization +Recyclability, Buy to fly ratio
+LCA of 3D printing machine
+Monitoring

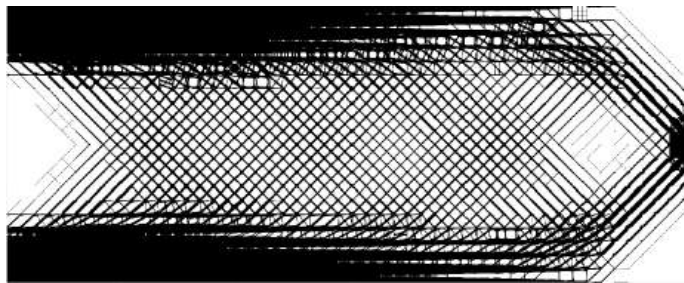


Mass minimization of parts

- Redesign through topology optimization
=> same performance but lower mass

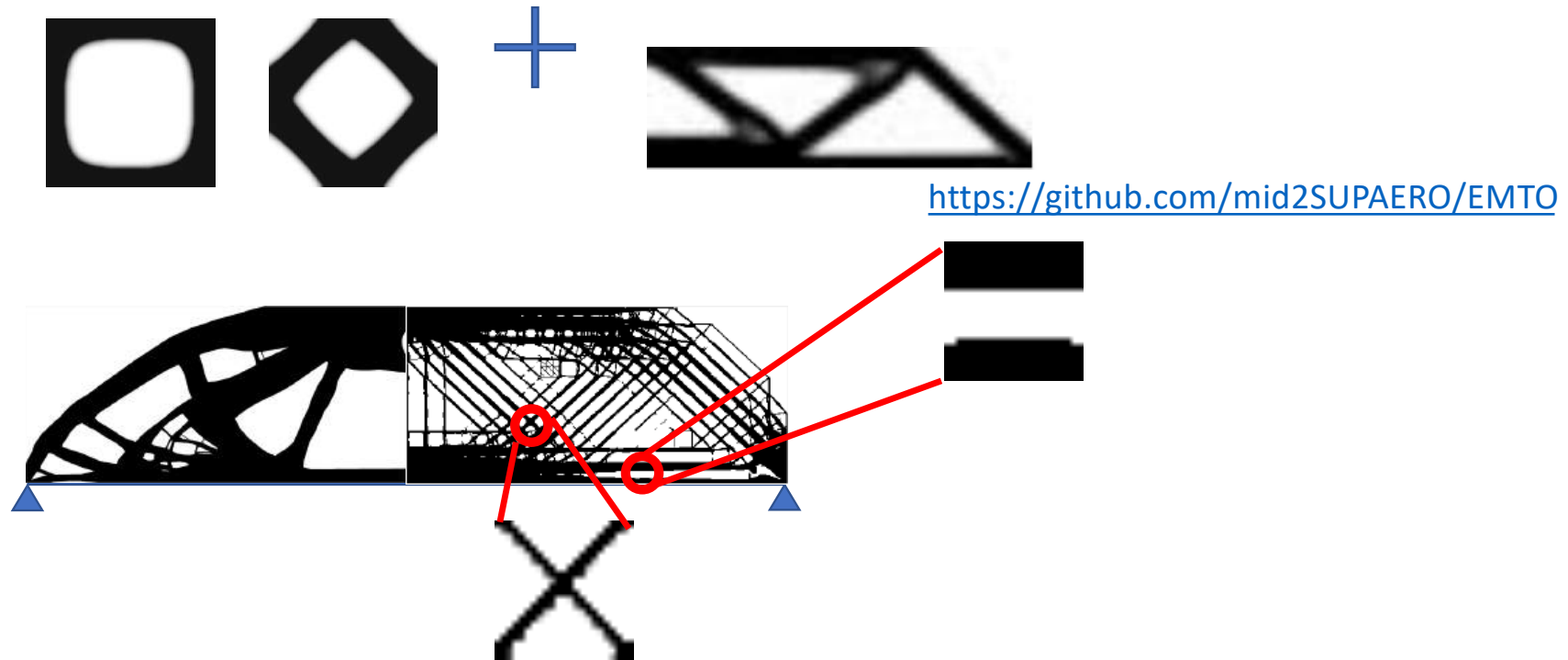


– One step further :
multiscale topology
optimization



Multi-scale TO (well connected+ locally-oriented)

A two level optimization that combines Unit cell design & Topology Optimization



Xia L, Breitkopf P (2015) Design of materials using topology optimization and energy-based homogenization approach in Matlab. Struct Multidisc Optim 52(6):1229–1241.

Wu, Jun, Ole Sigmund, and Jeroen P. Groen. "Topology optimization of multi-scale structures: a review." Structural and Multidisciplinary Optimization 63.3 (2021): 1455-1480.



Multiscale Topology Optimization

Macroscale Problem

need to solve



$$\text{minimize}_{x_{\text{dens}}^i, x_a^i, x_b^i, \dots} c = u^T K u$$

$$\text{subject to } K u = f$$

$$\sum_{i=1}^n \sum_{j=1}^m \rho_{ij} \leq n \times m \times v_f$$

$$\epsilon < \rho_{ij} < 1$$

$$x^i = [x_{\text{dens}}^i, x_{\text{or}}^i, x_{\text{cub}}^i]$$

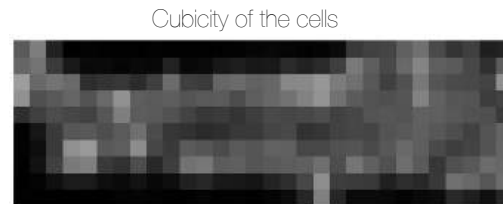
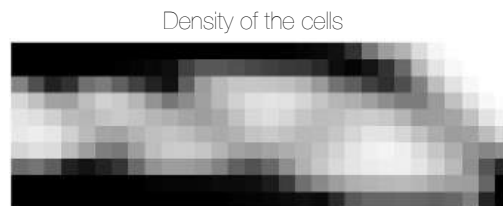
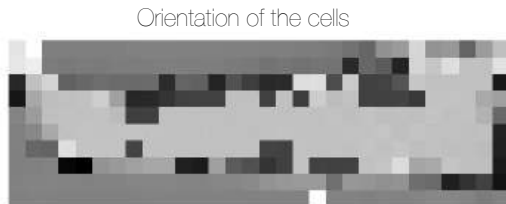
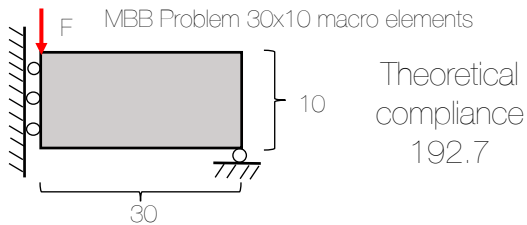
best unit cell per quad

Microscale Problem Unit cell with 3 properties
macro-density, angle, cubicity

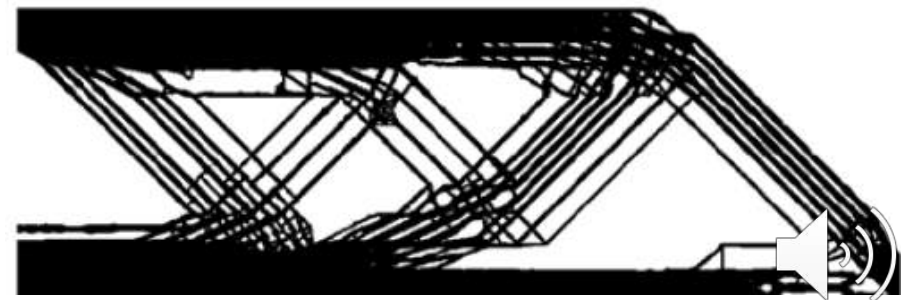
$$\text{minimize}_{\rho_{ij}} c_i = E_{1111}^{i\alpha} \times \left(1 - \frac{x_{\text{cub}}^i}{2}\right) + E_{2222}^{i\alpha} \times \frac{x_{\text{cub}}^i}{2}$$

$$\text{subject to } K_i u_i^{A(pq)} = f_i^{(pq)}$$

$$\sum_{j=1}^m \rho_{ij} \leq m \times x_{\text{dens}}^i$$



Since the objective is to create micro-structure with optimal properties towards specific directions, the objective function is a weighted function of the two components $E\alpha_{1111}$ and $E\alpha_{2222}$



Comparison

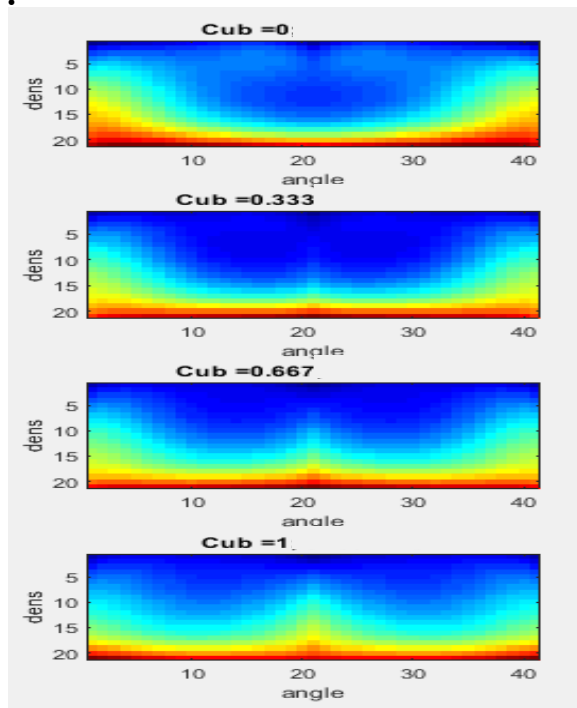
- Top88 versus EMTO



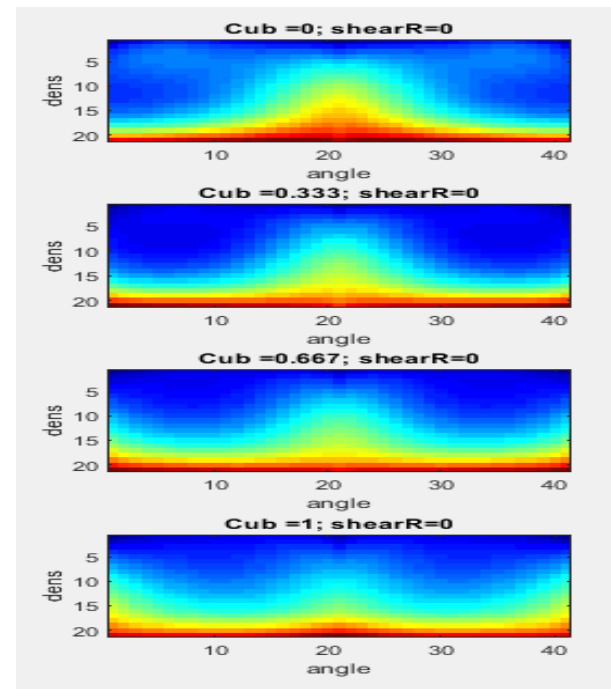
Elastic tensor's surrogate

to Avoid Nelement*cellOptimization at each macroOptimization step

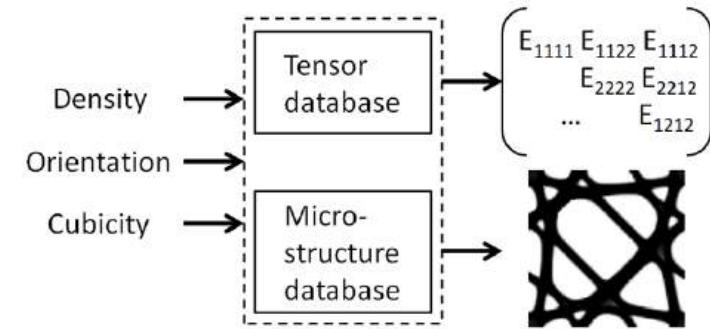
- 3 inputs : macro-density, angle, cubicity
- 6 outputs : elastic tensor values
- Gaussian interpolation : capture local effects but mitigate noise
- E1111:



E2222:



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23

Efficient Multiscale Topology Optimization

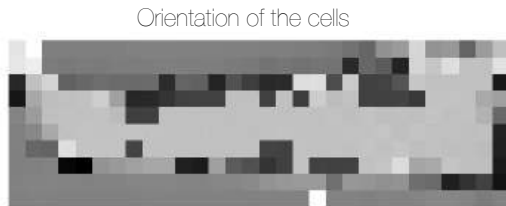
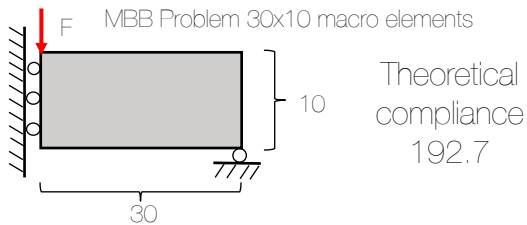
Macroscale Problem

$$\underset{x_{\text{dens}}^i, x_a^i, x_b^i, \dots}{\text{minimize}} \quad c = u^T K u$$

$$\text{subject to} \quad K u = f$$

$$\sum_{i=1}^n \sum_{j=1}^m \rho_{ij} \leq n \times m \times v_f$$

$$\epsilon < \rho_{ij} < 1$$



$$\longrightarrow x^i = [x_{\text{dens}}^i, x_{\text{or}}^i, x_{\text{cub}}^i] \longrightarrow$$

Nearest Optimal

$$x^i = [x_{\text{dens}}^i, x_{\text{or}}^i, x_{\text{cub}}^i]$$

Density of the cells



Cubicity of the cells



Gaussian Process Regression

$$\mathbf{E}_{\text{pred}}(x^i) = \frac{\sum_{l=1}^k G(x^i, x_l) \mathbf{E}_{\text{db}}(x_l)}{\sum_{l=1}^k G(x^i, x_l)}$$

$$G(x^i, x_l) = \exp\left(\frac{-d_{\text{eucl}}(x^i, x_l)^2}{2b^2}\right)$$

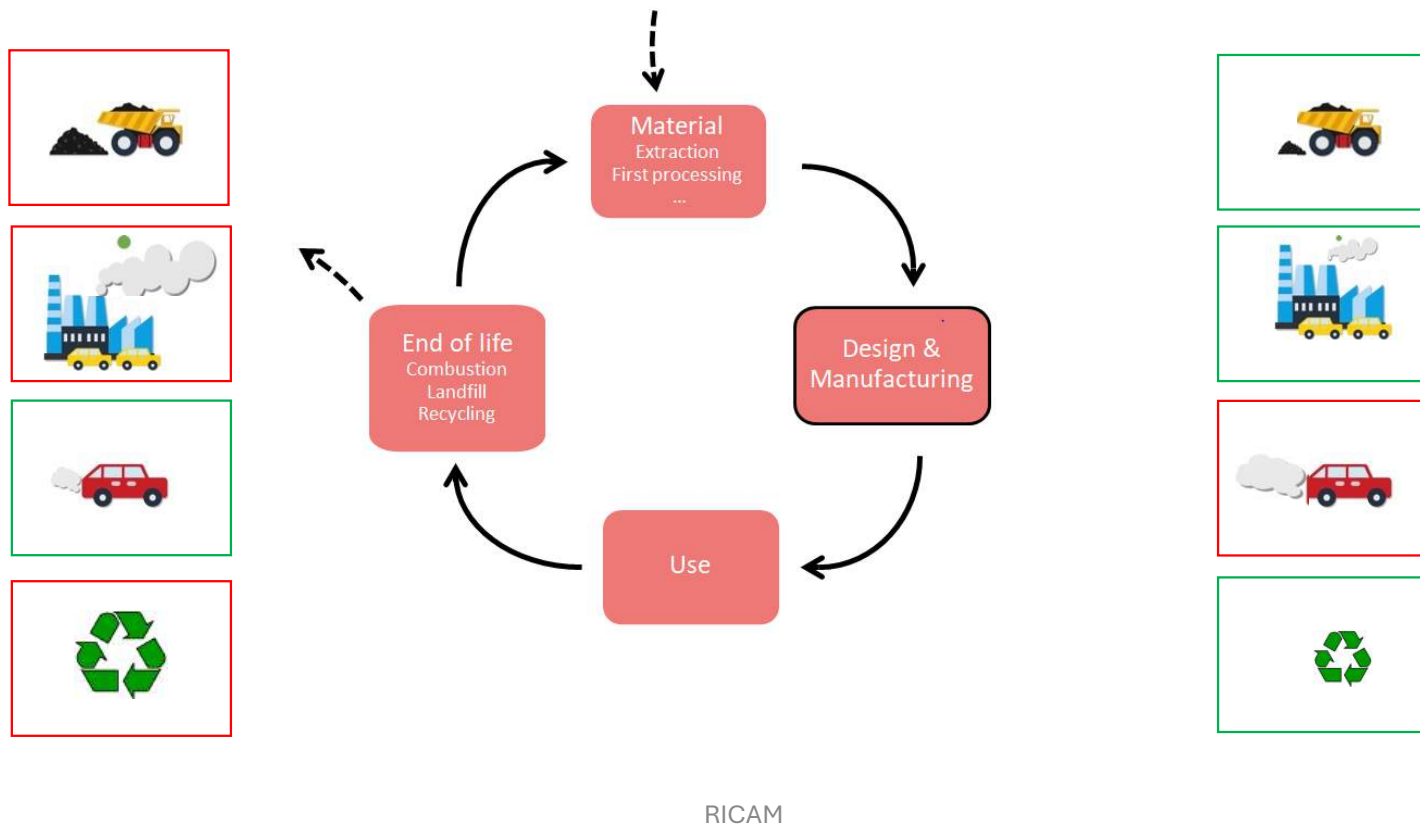
$$x^{i'} = [x_{\text{dens}}^i + \Delta, x_{\text{or}}^i, x_{\text{cub}}^i] \quad \Delta = 0.01$$

$$\frac{\partial \mathbf{E}_{\text{pred}}}{\partial x_{\text{dens}}}(x^i) \approx \frac{\mathbf{E}_{\text{pred}}(x^{i'}) - \mathbf{E}_{\text{pred}}(x^i)}{\Delta}$$



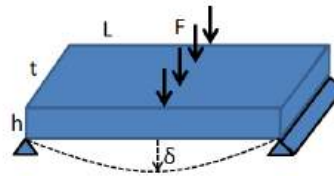
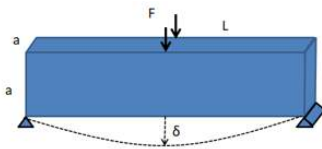
Hypothesis 2

- CO_2 emissions minimization of parts
 - If material choice is **free** => more complicated

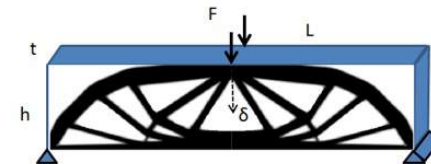


First zoom

Missing point from Ashby's theory: The absence of a simple analytical relation between compliance and volume fraction.



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Procedia CIRP
Volume 109, 2022, Pages 454-459



Ecodesign with topology optimization

Edouard Duriez ^a, Joseph Morlier ^a, Catherine Azzaro-Pantel ^b, Miguel Charlotte ^a

Show more

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Thanks

<https://doi.org/10.1016/j.procir.2022.05.278>

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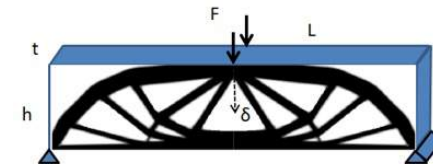
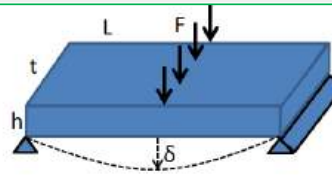
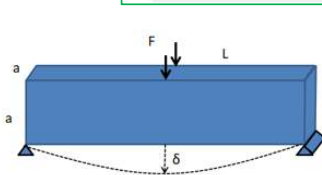
Get rights and content

Open access



- # Material index
- If fixed material and process,
 CO_2 minimization = mass minimization
 - Material choice through indices introduced by Ashby
 => uncouple material choice and part sizing
 - Include the geometrical design ($\mathcal{D}$) in the variables :

Properties	Bending beam (Ashby)	Bending plate (Ashby)	Duriez et al. (2022b)
Free variables	a, m	h, m	$t, \mathcal{D}, m$
Fixed	$L, \mathcal{D}$	$L, t, \mathcal{D}$	$L_{\max}, h_{\max}$
Constraint	$\delta_{\max}$	$\delta_{\max}$	$\delta_{\max}$



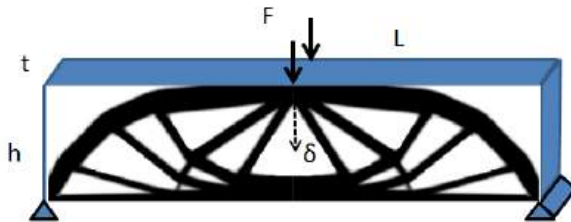
Ashby, M.F., 2004. Materials selection in mechanical design. 2. ed., reprinted ed., Elsevier Butterworth-Heinemann, Amsterdam.

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Deriving the material index

- Problem considered:



$$\arg \min_{mat, \mathcal{D}, t} CO_2^{tot}(mat, \mathcal{D}, t)$$

$$s.t. \quad \delta \leq \delta_{max}$$

$$mat = \{E, \rho, CO_{2mat}^i\} \in \Phi$$

$$0 < v_f(\mathcal{D}) \leq 1$$

- Objective function:

$$CO_2^{tot} = CO_2^{mat} \times M + CO_2^{veh} \times LD \times M$$

How many miles does an airplane like a 777 fly over the course of its lifetime?

777: A 30-year lifetime. 3,500 hours a year as an average. An average speed of 500 miles per hour. $30 \times 3500 \times 500 = 52,500,000$ miles i.e. LD.

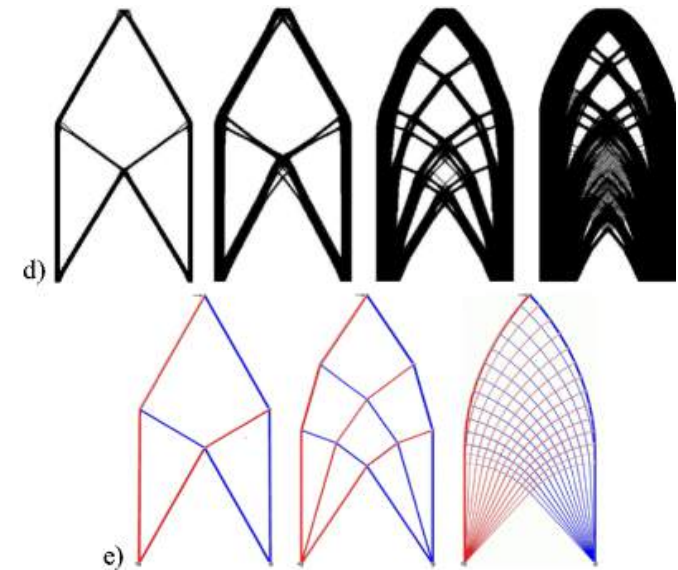
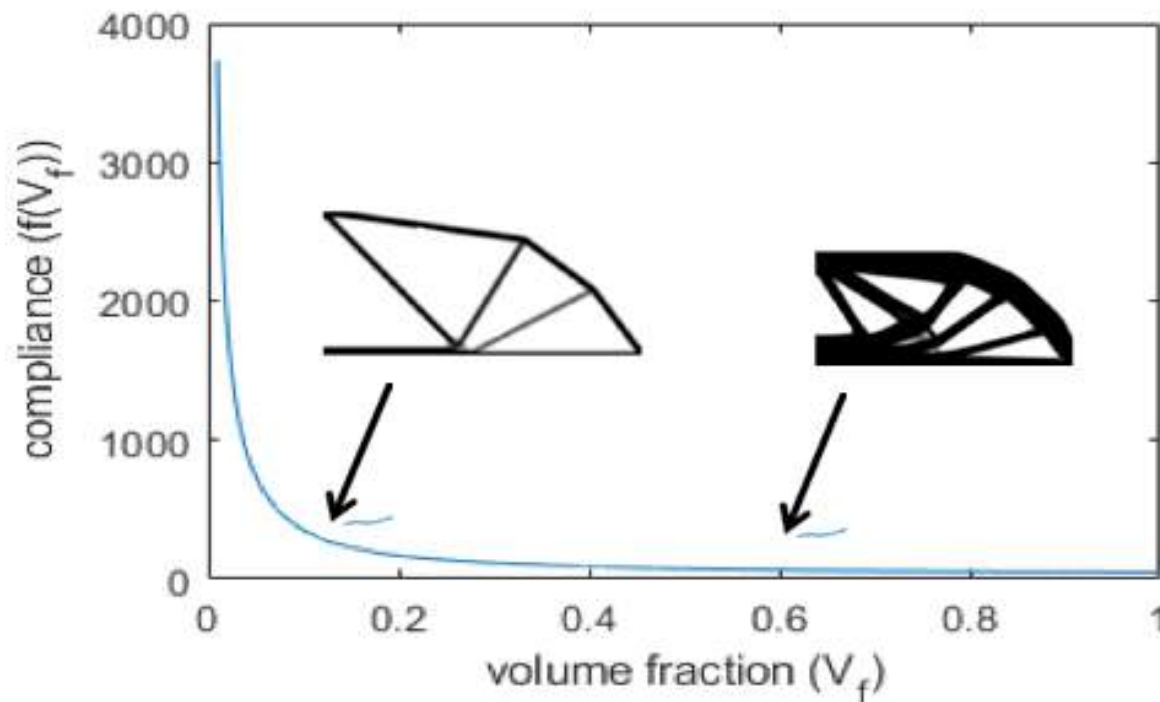


Topology optimization pareto front

V_f : volume fraction (ratio of space containing material)

$f(V_f)$: compliance – volume fraction pareto front

Edouard Duriez, Miguel Charlotte, Catherine Azzaro-Pantel et al. On some properties of the compliance-volume fraction Pareto front in topology optimization useful for material selection., 27 December 2022, PREPRINT (Version 1) available at Research Square [<https://doi.org/10.21203/rs.3.rs-2390440/v1>]

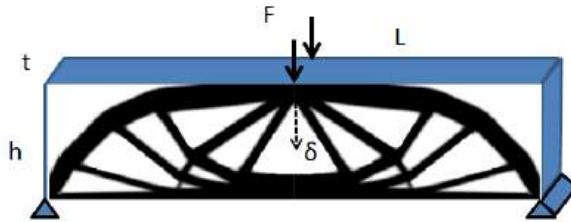


Sigmund, O., Aage, N., & Andreassen, E. (2016). On the (non-) optimality of Michell structures. Structural and Multidisciplinary Optimization, 54, 361-373.



Deriving the material index

- Problem considered:



$$\arg \min_{mat, \mathcal{D}, t} CO_2^{tot}(mat, \mathcal{D}, t)$$

$$s.t. \quad \delta \leq \delta_{max}$$

$$mat = \{E, \rho, CO_{2mat}^i\} \in \Phi$$

$$0 < v_f(\mathcal{D}) \leq 1$$

- Objective function:

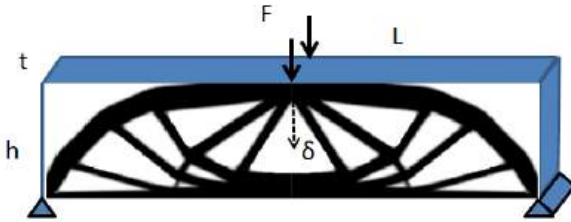
$$CO_2^{tot} = CO_2^{mat} \times M + CO_2^{veh} \times LD \times M$$



Deriving the material index

starting from $C = U^T K U = U^T F = U(i_F) F(i_F) = F U(i_F)$
 $M = \rho t L h V_f$

- Problem considered:



compliance

$$C \leq F \delta_{max}$$

$$\frac{f(V_f) F}{t E} = \delta_{max}$$

If t is a free variable, it can be chosen as in compliance to achieve the minimum mass

$$t = \frac{f(V_f) F}{\delta_{max} E} \quad \text{thus} \quad M = \frac{L h F}{\delta_{max}} \frac{\rho}{E} f(V_f) V_f$$

$$CO_2^{tot} = (CO_2^{mat} + L D CO_2^{veh}) \times \frac{\rho}{E} \frac{L h F}{\delta_{max}} f(V_f) V_f$$

Material $f_3(M)$

functional $f_1(F)$

topology index $f_2(G)$



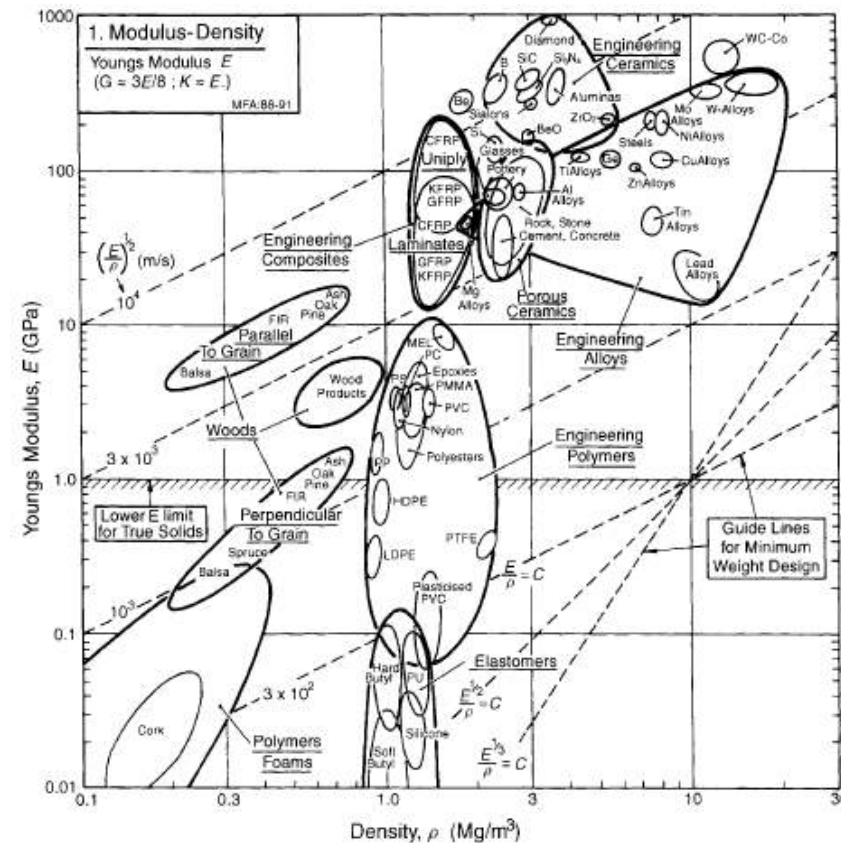
REMINDER !!

- Many materials
 - Competing properties
 - => Ashby indexes: $f_3(M)$
- $$P = f_1(F) \times f_2(G) \times f_3(M)$$

F: Functional constraints

G: Geometrical constraints

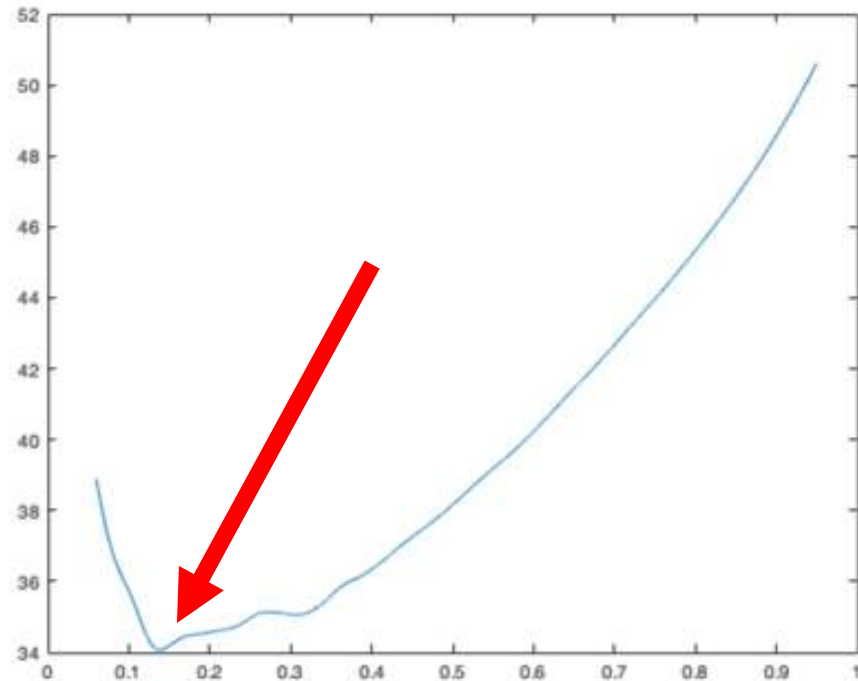
M: Material properties



M. F. Ashby et Kara Johnson. « Materials and Design : The Art and Science of Material Selection in Product Design ». In : 2002.

Minimize topology index $f_2(G)$

- Topology index : $f(V_f)V_f$:
Minimize this to minimize mass
- Thickness t is finally adjusted to satisfy constraint



Data for Material index

functional improvement can be found in lightweight components for transport systems

Table 8. **Fuel** consumption reduction coefficients for different vehicle types and related life time impact savings per kg of weight reduction.

Transport system	Energy source	FRC [26]	Service life	Eco-Impact (ReCiPe H/A)	Life time savings (ReCiPe H/A)	Equivalent electrical energy
Gasoline car	Gasoline	0.5 l / (100kg*100km)	200000km	0.121 Pts/l	1.21 Pts/kg	85 MJ
Diesel car	Diesel	0.24 l / (100kg*100km)	200000km	0.141 Pts/l	0.68 Pts/kg	48 MJ
Short distance train	Electricity	300 kJ / (1000kg*km)	3.5*10 ⁶ km	0.051 Pts/kWh	14.88 Pts/kg	1050 MJ
Long distance train	Electricity	100 kJ / (1000kg*km)	10*10 ⁶ km	0.051 Pts/kWh	14.17 Pts/kg	1000 MJ
Short distance aircraft	Kerosene	12.5 ton / (100kg*year)	25 year	0.134 Pts/l	335 Pts/kg	23647 MJ
Long distance aircraft	Kerosene	103 ton / (100kg*year)	25 year	0.134 Pts/l	2760 Pts/kg	194852 MJ

Kellens, Karel, et al. "Environmental impact of additive manufacturing processes: does AM contribute to a more sustainable way of part manufacturing?." *Procedia Cirp* 61 (2017): 582-587.



Results

- Results change depending on the application:

- Aircraft

Material	E (GPa)	ρ (kg/m ³)	CO_{2mat}^i (kgCO ₂ /kg)	Idx (kgCO ₂ /N/m)
Al alloy	70.8	2795	13.0	3.90×10^{-3}
Stainless steel	197	7915	6.15	3.97×10^{-3}
Ti alloy	115	4425	40.4	3.80×10^{-3}
Inconel 713	205	7900	15.5	3.81×10^{-3}

mass : 5,24kg

5.2445

CO₂ emissions:
517 tons

1.0e+05 * 5.1744

- bridge

Material	E (GPa)	ρ (kg/m ³)	CO_{2mat}^i (kgCO ₂ /kg)	Idx (kgCO ₂ /N/m)
Al alloy	70.8	2795	13.0	5.13×10^{-7}
Stainless steel	197	7915	6.15	2.47×10^{-7}
Ti alloy	115	4425	40.4	1.56×10^{-6}
Inconel 713	205	7900	15.5	5.97×10^{-7}

mass : 5,47kg

5.4761

CO₂ emissions:
33,67 kg

33.6783



To go deeper



A fast method of material, design and process eco-selection via topology optimization, for additive manufactured structures

Edouard Duriez^a  , Catherine Azzaro-Pantel^b, Joseph Morlier^a, Miguel Charlotte^a

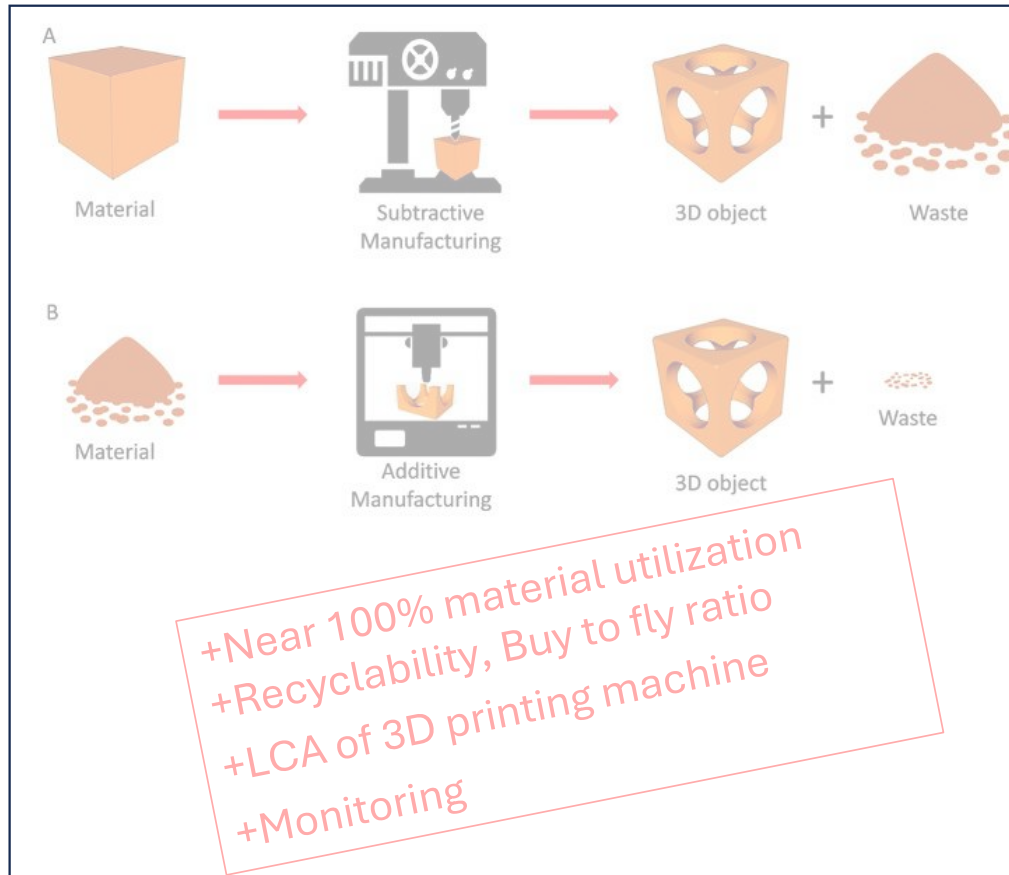
Properties	Bending beam (Ashby)	Bending plate (Ashby)	Duriez et al. (2022b)	Our problem
Free variables	a, m	h, m	$t, \mathcal{D}, m$	$\mathcal{D}, m, p$
Fixed	$L, \mathcal{D}$	$L, t, \mathcal{D}$	$L_{\max}, h_{\max}$	$L_{\max}, h_{\max}, t_{\max}$
Constraint	$\delta_{\max}$	$\delta_{\max}$	$\delta_{\max}$	$\delta_{\max}$

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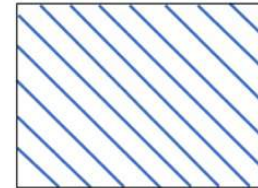
Process is AM, but WHY?

<https://dfam.substack.com/p/dfam-education-in-2022>



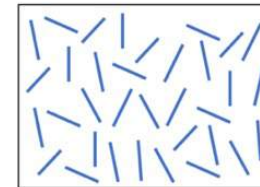
RICAM

Regular and periodic



Natural (optimal?)

Random



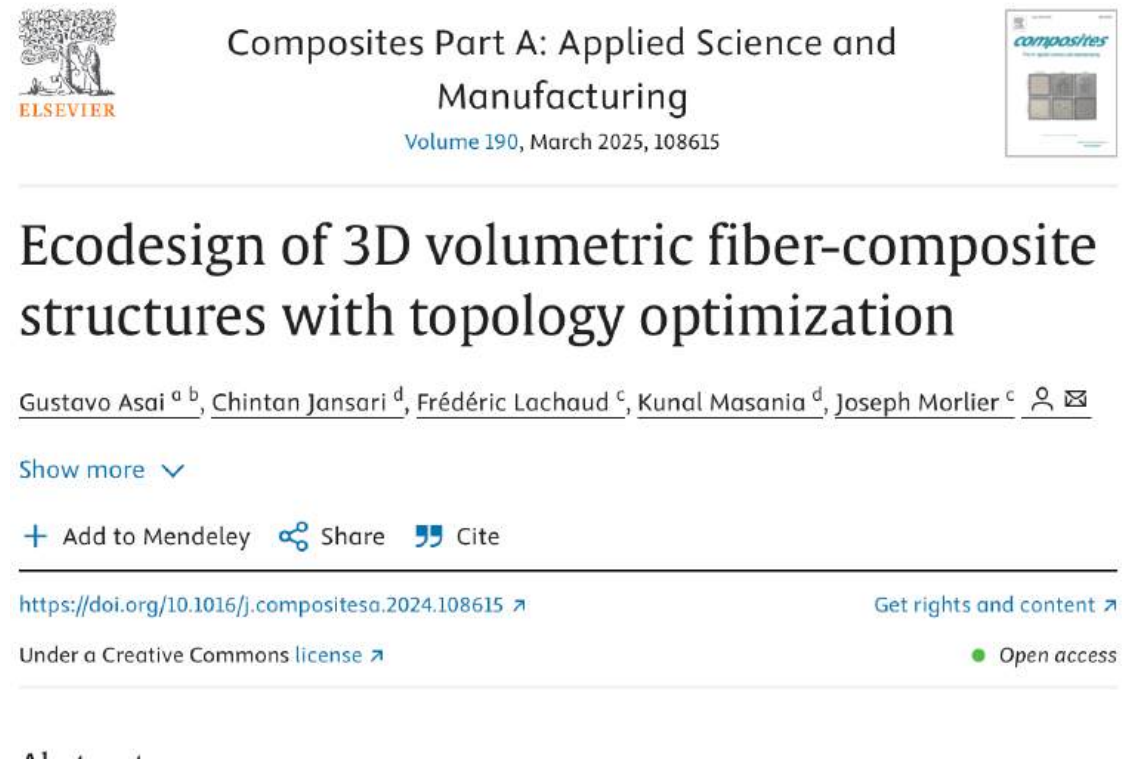
Non-periodic and specific (optimal)

+ Automatic Fiber Placement + eco-fiber/resin selection + Monitoring



Add Ecomaterial selection, printability

Results



https://github.com/mid2SUPAERO/SOMP_Ansv



Add Ecomaterial selection, printability

Results

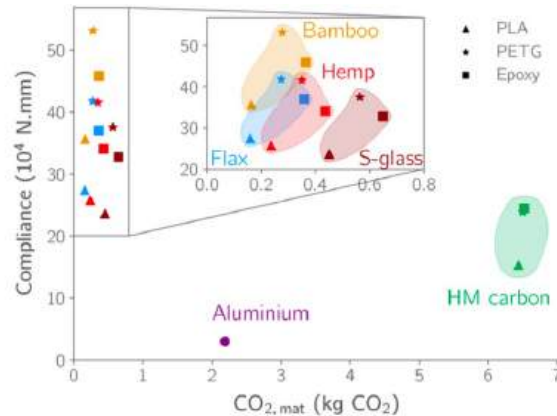


Figure 7: Compliance versus material production footprint of the optimal designs, grouped by fiber.

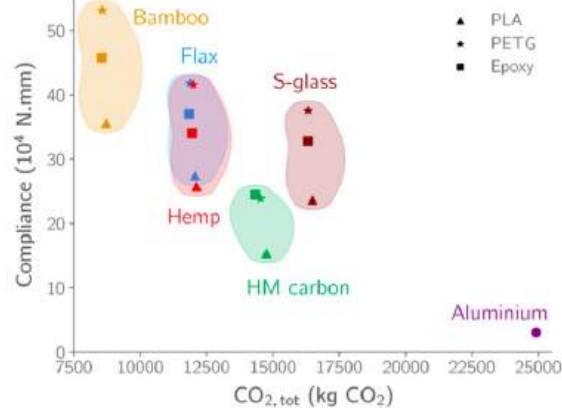


Figure 8: Compliance versus total footprint of the optimal designs, grouped by fiber.

RICAM

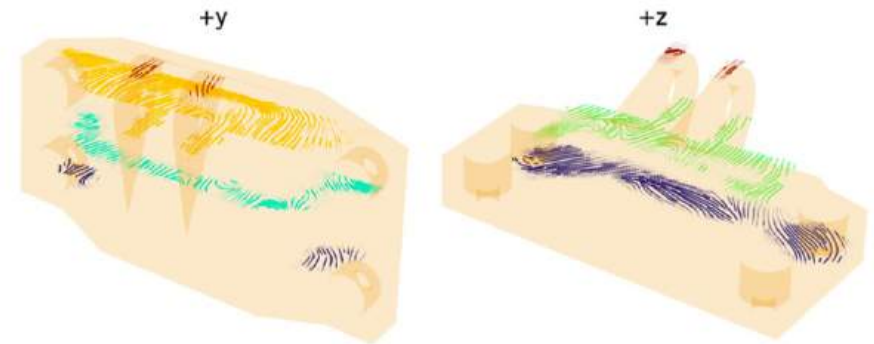


Figure 11: Examples of fiber distribution on slices of the optimal designs.

Random initial orientations
Compliance: 27.04×10^4 N.mm
Printability: 74.2%

Initial orientation: 0°
Compliance: 26.88×10^4 N.mm
Printability: 73.1%

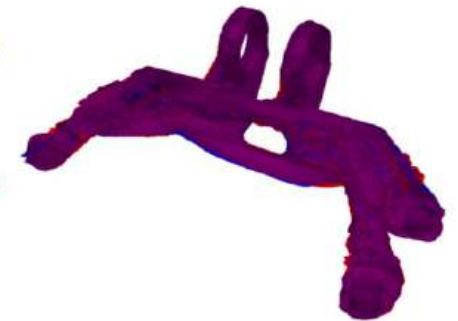


Figure 16: Isosurfaces of the optimal designs for random and 0° initial orientations.

https://github.com/mid2SUPAERO/SOMP_Ansv



Au programme

1. Problem $\mathcal{P}_1$, where the topology is fixed (truss) or parametric (wing),
2. Problem $\mathcal{P}_2$, where the topology is free.

3. On some coupled problems $\mathcal{P}_3$

4. Conclusions

Aerospace design is a coupled problem $\mathcal{P}_3$

- MDO implies to solve the coupling of physics (aerodynamic and structures)
- To limit the computational cost of high-fidelity simulations we extensively use surrogate modeling and bayesian optimization



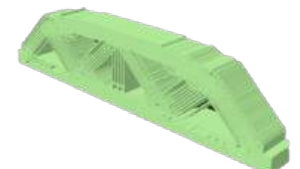
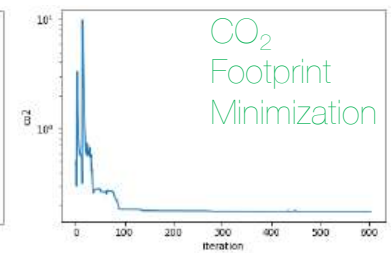
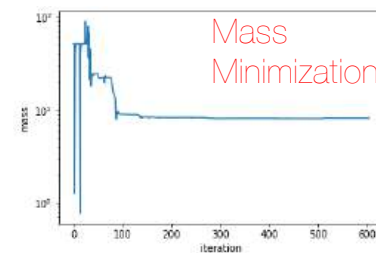
Aircraft Design

Fluid x control x physics x applied maths x structures & materials

In AD: Strong coupling
Between
Disciplines (MDO)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Avionics group	1	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Electrical group	2	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Escape system	3	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Armament	4	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Landing gear	5	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Hydraulics group	6	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Flight control system	7	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Environment and control system	8	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Power plant group	9	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Fatigue group	10	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Aero elastic group	11	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Stress group	12	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Materials group	13	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Fly wheel	14	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Empennage group	15	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Rear fuselage	16	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Fuselage group	17	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•

Topology x Material x Process



$$\text{Range} = V t_f = V \times \underbrace{\left(\frac{L}{D}\right)}_{\text{aircraft designer}} \times \underbrace{I_{sp}}_{\text{propulsion system designer}} \times \underbrace{\ln\left(\frac{W_i}{W_f}\right)}_{\text{structure designer}}$$

Testcase #1: Aircraft optimization

Topology is free in the MDO problem

f1=Minimize (-range) and f2=minimise (GWP) % second problem

$$f = \alpha * f1 + (1 - \alpha) * f2$$



<https://pymoo.org>



LCA4MDAO

<https://github.com/mid2SUPAERO/LCA4MDAO>



Hybrid Aircraft Problem (MDOLab)

- Hybridised King Air C90GT from [OpenConcept](#), built in *OpenMDAO* format
- Four disciplines:
 - Aero (wing geometry)
 - Propulsion (with hybrid system)
 - Structure
 - Trajectory simulation
- 6 variables converted into LCA database entries

Model parameter	Ecoinvent entry
Battery weight	battery cell production, Li-ion
Motor weight	electric motor production, vehicle
Engine weight	internal combustion engine production, passenger car
Empty weight	aluminium production, primary, ingot
Fuel used	market for kerosene
Electricity used	market group for electricity, low voltage

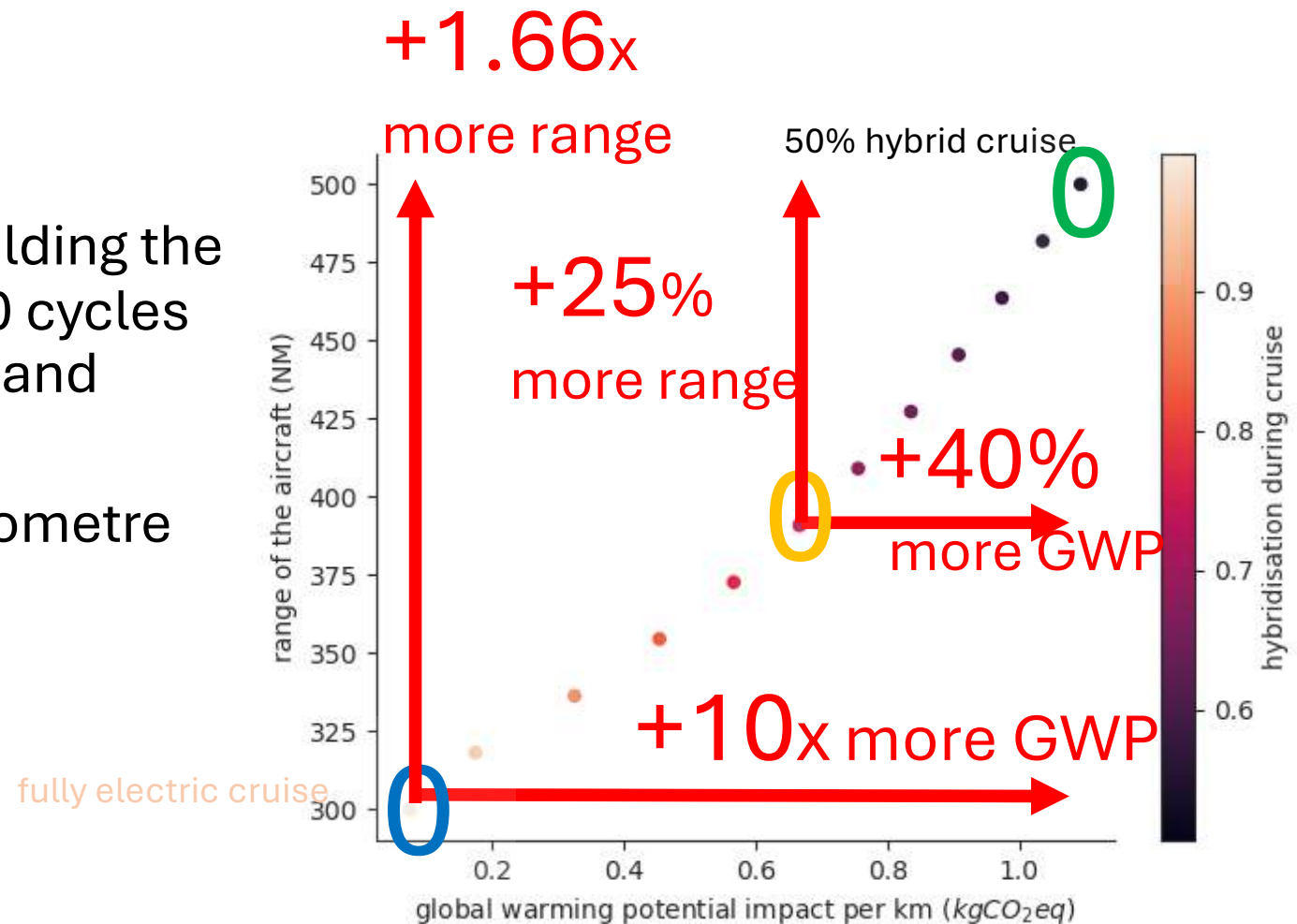
Benjamin J. Brelje and Joaquim R. R. A. Martins, "Development of a Conceptual Design Model for Aircraft Electric Propulsion with Efficient Gradients", 2018 AIAA/IEEE Electric Aircraft Technologies Symposium, AIAA Propulsion and Energy Forum, (AIAA 2018-4979) DOI: 10.2514/6.2018-4979

Eytan J. Adler and Joaquim R. R. A. Martins, "Efficient Aerostructural Wing Optimization Considering Mission Analysis", Journal of Aircraft, 2022. DOI: 10.2514/1.c037096



Results MOO

- LCA scope include building the aircraft and flying 1000 cycles at max range with fuel and electricity
- Functional unit is a kilometre flown

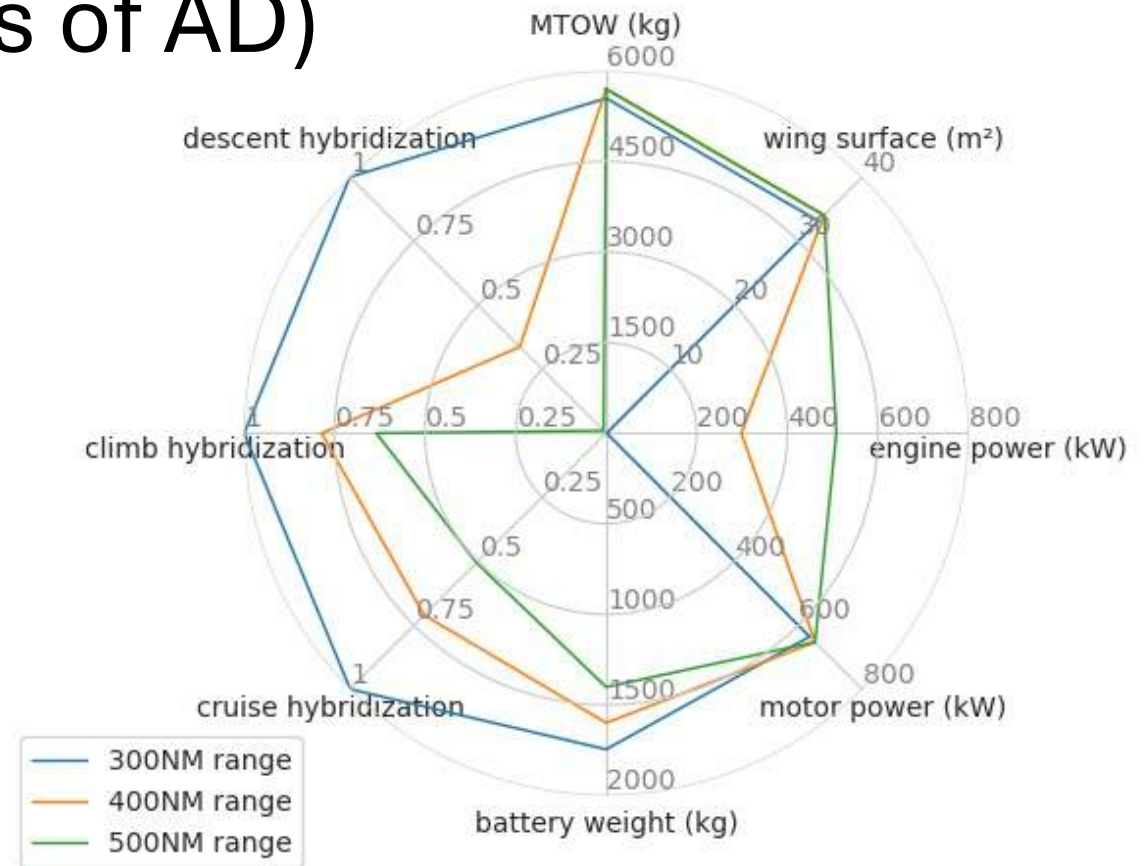


Results (link to physics of AD)

For the design variables,
reducing the range:

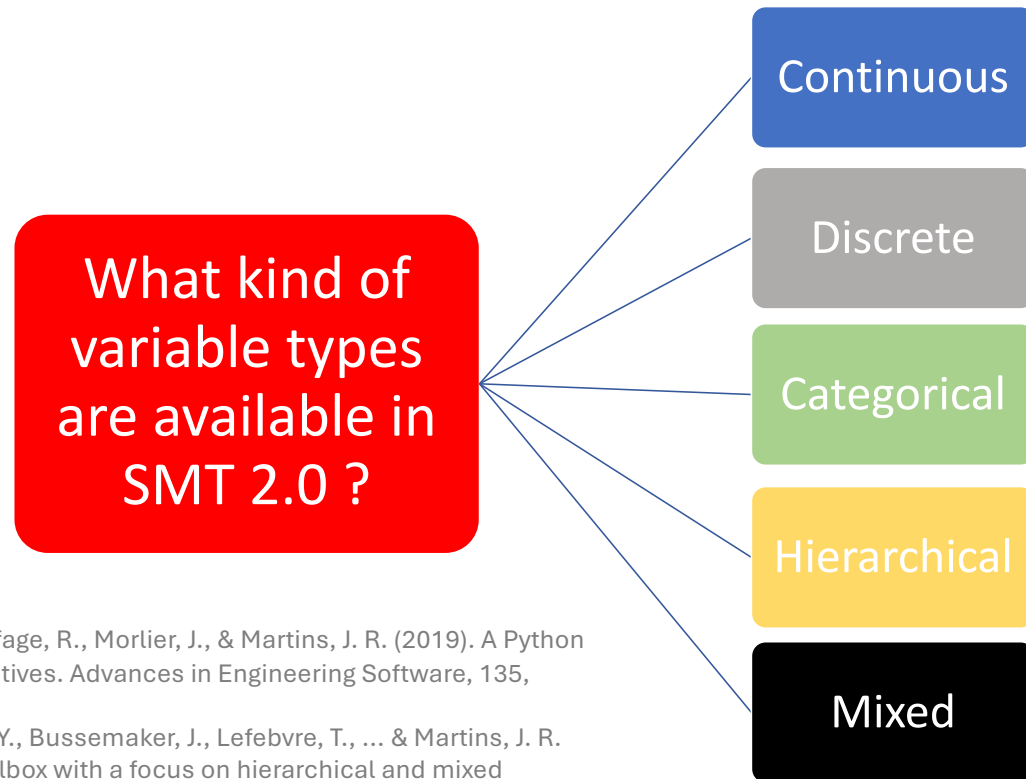
- increases the hybridization
- reduces the engine size
- increases the battery weight

variable	value	units
MTOW	5700	kg
wing surface	34	m ²
engine power	298	kW
motor power	652	kW
battery weight	1607	kg
fuel capacity	500	kg
cruise hybridisation	0.71	
climb hybridisation	0.785	
descent hybridisation	0.337	
GWP	0.712	kgCO ₂ eq/km



Another solution... Another story

- *material selection as hybrid/categorical variables ?*



Bouhlef, M. A., Hwang, J. T., Bartoli, N., Lafage, R., Morlier, J., & Martins, J. R. (2019). A Python surrogate modeling framework with derivatives. *Advances in Engineering Software*, 135, 102662.

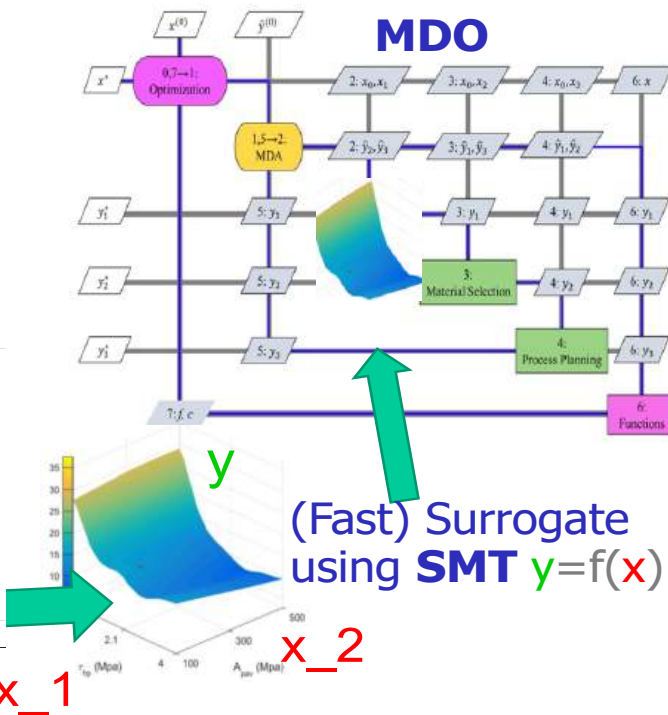
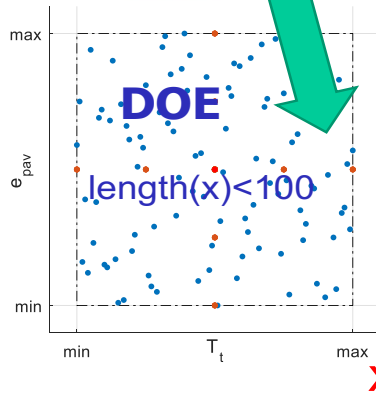
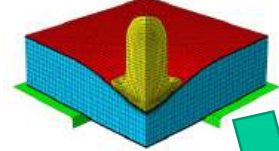
Saves, P., Lafage, R., Bartoli, N., Diouane, Y., Bussemaker, J., Lefebvre, T., ... & Martins, J. R. (2024). SMT 2.0: A Surrogate Modeling Toolbox with a focus on hierarchical and mixed variables Gaussian processes. *Advances in Engineering Software*, 188, 103571.



Interested in Surrogate modeling? GP/BO?

Big picture?

(Expensive)
simulation



(Fast) Surrogate
using **SMT** $y=f(x)$



<https://github.com/SMTorg/smt>



Au programme

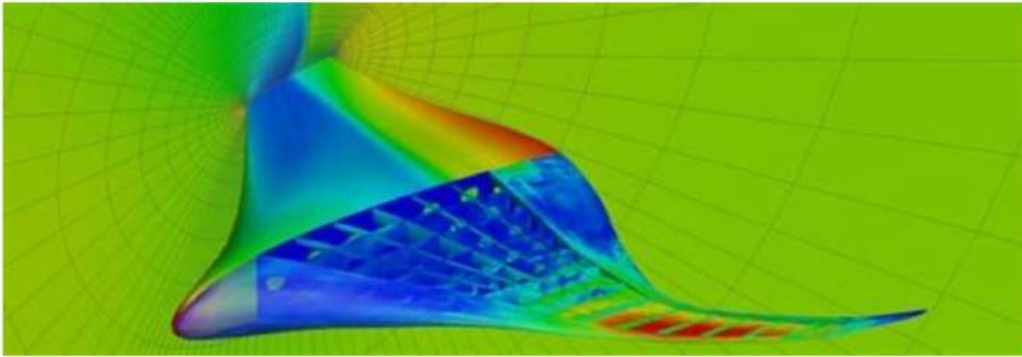
1. Problem $\mathcal{P}_1$, where the topology is fixed (truss) or imposed (wing),
2. Problem $\mathcal{P}_2$, where the topology is free.
3. On some coupled problems $\mathcal{P}_3$

4. Conclusions

Conclusions

- Proof of concept of greener structures
- VAE, Surrogate Optimization can help for Design acceleration especially when material choice is in the loop





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<http://mdolab.engin.umich.edu>

Optimization [MDO] for connecting people?

Publié le 14 février 2019

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joseph morlier

Professor in Structural and Multidisciplinary Design Optimization, ... any idea?
2 articles

74 31 3 0

<https://www.linkedin.com/pulse/optimization-mdo-connecting-people-joseph-morlier/>



https://www.tripadvisor.fr/LocationPhotoDirectLink-g187529-d574612-i349532022-Museum_of_Natural_Science_Museo_de_Ciencias_Naturales-Valencia_Province_o.html

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Publié le 17 février 2020

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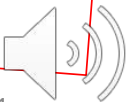
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Professor in Structural and Multidisciplinary Design Optimization, ... any idea?

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RICAM



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